

AI Smart Device Control & Automation Assistant

STEP 04 PROJECT DOCUMENTATION — Laptop Automation Foundation

Item	Current Project Details
Completed Steps	01 — Initialization; 02 — Configuration & Command Model; 03 — Command Parser; 04 — Laptop Automation
Platform	Windows / PowerShell
Python	3.11.9 — venv
pytest	9.1.1
Automated Tests	8/8 PASS — 100%
Manual Test	Notepad launch — PASS
Status	COMPLETED — PASS

1. Executive Summary

Step 04 introduced the laptop automation foundation. A dedicated **LaptopController** executes a small predefined allowlist of Windows applications. It validates the device, action, target, and allowed application before execution. The complete test suite passed with **8/8 tests in 0.09 seconds**. A manual functional test also successfully launched Windows Notepad.

2. Laptop Automation Architecture

User Command ↓ Command Parser ↓ DeviceCommand ↓ LaptopController ↓ Allowlisted Application ↓ Windows Application

The parser interprets the command while the controller performs the actual laptop action. This keeps command interpretation separate from execution.

3. Laptop Controller

The controller provides `open_application()` and `execute()`. The initial allowlist contains Windows Notepad and Calculator.

Application	Executable	Status
notepad	notepad.exe	Enabled
calculator	calc.exe	Enabled

4. Execution Safety Boundary

The controller does not execute arbitrary shell commands. It checks the requested application against a predefined allowlist and uses `subprocess.Popen(..., shell=False)`. Non-laptop commands, unsupported actions, missing targets, and non-allowlisted applications are rejected.

5. Parser Integration

Existing Step 03 parser behavior was preserved while adding Notepad and Calculator recognition.

Example	Structured Result
Open Chrome	laptop → open → chrome
Open VSCode	laptop → open → vscode
Open Notepad	laptop → open → notepad
Open Calculator	laptop → open → calculator

6. Automated Testing — Final Result

The complete project test suite was executed from the project root. All five Step 03 parser tests and all three Step 04 controller tests passed.

Test	Result
test_turn_on_light	PASSED
test_turn_off_fan	PASSED
test_open_chrome	PASSED
test_bluetooth_devices	PASSED
test_unknown_command	PASSED
test_laptop_controller_initialization	PASSED
test_non_laptop_command_is_rejected	PASSED
test_unsupported_application_is_rejected	PASSED
Total	8 passed in 0.09s

7. Manual Functional Validation

A direct Python interactive test passed a DeviceCommand targeting Notepad to LaptopController.execute(). The controller returned **True** and printed the successful launch message. The user confirmed that Notepad opened successfully. The command was executed twice and both calls returned True.

```
controller.execute(command) → ■ Application launched: notepad → True
```

8. Troubleshooting / Lessons Learned

Issue	Resolution / Lesson
Chrome parser regression	Restored the existing Chrome rule while adding new laptop application rules; existing functionality must be preserved.
Package imports	Retained the src package/import structure established in Step 03.
Execution safety	Used an explicit application allowlist instead of arbitrary command execution.
Functional verification	Separated automated validation from the real Notepad launch test.

9. Step 04 Milestone Status

Milestone	Status
LaptopController created	COMPLETED
Application allowlist	COMPLETED
Laptop command validation	COMPLETED
Unsupported command rejection	COMPLETED
Automated tests	8/8 PASS — 100%
Real Notepad launch	PASS

10. Project Status After Step 04

Step	Milestone	Status
01	Project Initialization & Environment Setup	COMPLETED
02	Configuration & Device Command Model	COMPLETED
03	Command Parser / Intent Model	COMPLETED — 5/5 PASS
04	Laptop Automation Foundation	COMPLETED — 8/8 PASS + Notepad PASS
05	Bluetooth Device Discovery	Pending
06	Bluetooth Device Control	Pending
07	IoT Control Foundation	Pending
08	AI Intent Integration	Pending
09	Voice Input Integration	Pending
10	Voice Response Integration	Pending
11	Command Router	Pending
12	Activity Logging	Pending
13	Error Handling	Pending
14	Dashboard Foundation	Pending
15	Dashboard Integration	Pending
16	Automated Testing	Pending — final suite later
17	End-to-End Testing	Pending
18	Performance Evaluation	Pending
19	Documentation & Final Report	Pending
20	Final Demonstration & Closure	Pending

11. Step 04 Conclusion

STEP 04 — LAPTOP AUTOMATION FOUNDATION: PASS. The project now has a controlled laptop execution layer. The complete test suite passed with 8/8 tests in 0.09 seconds, and a manual functional test successfully launched Windows Notepad through LaptopController.

12. Revision Record

Revision	Update
STEP 01	Project initialization, environment setup, validation, roadmap, and documentation foundation.
STEP 02	Configuration, configuration loader, DeviceCommand model, validation, and milestone status.
STEP 03	Deterministic command parser, intent mappings, unknown-command handling, package/import correction, and 5/5 PASS.
STEP 04	Added LaptopController, application allowlist, parser integration, 8/8 automated tests PASS in 0.09s, and successful manual Notepad launch.
Next	Continue with Step 05 — Bluetooth Device Discovery and update this same report after validation.